卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-SG-0403 30-B003701S-S0403

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CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 ±660kV Lahore Converter Station 工程 PROJECT　施工图 Detail Design 阶段 DESIGN STAGE

水布 Hydraulic 专业 SPECIALITY　第 04 卷 VOLUME　第 03 册 BOOK

卷册名称 Volume name　站区工业水管道布置图 Layout Drawing Of Outdoor Industrial Water Piping

批　准 APPROVED BY	
审　核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	罗
图 纸 DRAWING 3 张 NOS.	打印件 PRINTED COPIES 0 本 NOS.

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2	30-B003701S-S0403-002	站区工业水管道平面布置图	A	1		
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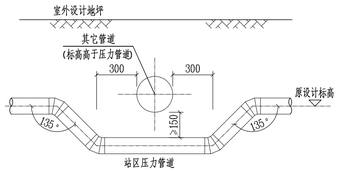
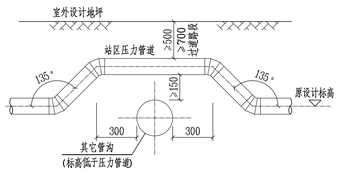
3	30-B003701S-S0403-003	站区工业水管材料表	A	1		
		Bill Of Material For Industrial Water Piping				

注:

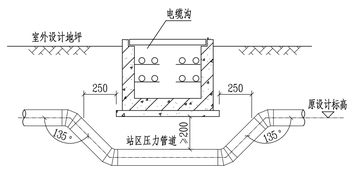
站区压力管道施工说明



穿越管道下弯典型图



穿越管道上弯典型图



穿越电缆沟下弯典型图

注：

1、以上压力管道为站区补给水管、工业水管和生活水管。

2、给水管与管沟交叉原则：

1）让电缆沟。2）让重力流下水管道。3）小管径让大管径的管道。

3、在道路范围内，给水管应从所穿管道下方穿过；在场地范围内，如给水管需从管道上方穿越，其管顶至地面不应小于500mm。

1、本工程座标系统采用站区座标系统，站区座标系统与巴基斯坦UTM-43N平面坐标系的换算公式详见土建相关卷册。

2、本图卷册标高采用EGM2008高程系统，换流变区域硬化地坪标高为193.60m，其它区域地坪标高为193.40m，详见土建相关卷册。

3. 平面座标以m为单位，尺寸以mm为单位，高程以m为单位。所注管道标高单位以m计，压力管道所注管道标高均为管中心标高。

4. 本工程补给水、工业、生活水源由打井抽水提供，从给水处理装置区域产水管接出。本工程设置1500m^{3}工业水池两座，综合泵房一座，综合泵房内设工业水泵、生活水供水设备、生活水箱。

5. 本册图中管道均采用直埋敷设，所有管道应敷设在100mm厚中粗砂垫层上，安装完毕回填前应进行严密性水压试验，水压试验合格后，需先修补防腐涂层，再覆土回填。厂区工业水管工作压力为0.55MPa，试验压力为1.05MPa，具体试验要求详见《给水排水管道工程施工及验收规范》（GB50268-2008）。

6. 管道原则上原土回填，但不能含有石块、砖头等硬物，并应在两侧同时分层夯实，回填砂的干容重应保证1.60t/m^3，回填土的干容重应保证1.45t/m^3，密实度0.93。

7. 道路地面荷载按汽-15级（加重双车并列）设计。

8. 在整个安装过程中，均应注意不允许管槽长期积水，并采取排施防止浮管。

9. 站区工业水管、生活水管、补给水管采用衬塑镀锌钢管，丝扣连接，管道与阀门之间采用法兰连接。

10. 钢管防腐等基本要求如下：

a. 表面预处理：钢管表面应无油污、泥土、盐份、潮气等杂物，用喷砂或人工方式彻底去除表面的氧化皮、锈，软明显地露出金属光泽，达到ISO8501-1标准Sa2.5或St3级，然后立即进行防腐涂料的涂装工作。此外钢管表面还需具有一定的表面粗糙性，粗糙度应在32.5~75μm之间。

b. 涂装环境：涂装前，除了前道涂层的表面要清洁、干燥外，还需注意底材温度应高于露点温度3℃，以上相对湿度应低于85%，以免管材的凝露影响涂料的附着力；环境的持续温度应高于5℃，保证固化反应。

部位	品种	油漆名称	道数	每道漆膜厚度μm
埋地钢管（镀锌钢材）外壁	封闭层	H53-42环氧封闭涂料	1	30
	中间层漆	842环氧云铁防锈漆	1	60
	面漆	546超厚浆型环氧沥青防锈漆	2	125

涂料厚度及涂装方法参见涂料厂家的有关说明要求。要求涂刷均匀，不得漏涂。需焊接管道两端各留裸管150mm左右，以便焊接。

11. 本工程图中未注明的管道交叉方式参见左图，现场可根据需要自行避让调整，上下转角应尽量采用45°弯头，管件材料均计入管长。

12. 压力管道要求遵循规范标准如下：

1）《给水衬塑符合钢管》 CJ/T 136-2007

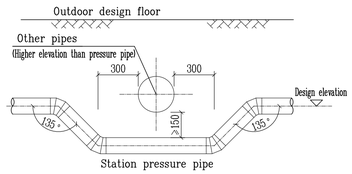
2）《给水衬塑可锻铸铁管件》 CJ/T 137-2008

3）《钢结构工程施工质量验收规范》 GB50205-2001

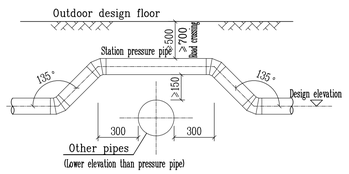
4）《电力建设施工及验收技术规范 火力发电厂焊接篇》 DL869-2012

5）《给水排水管道工程施工及验收规范》 GB50268-2008

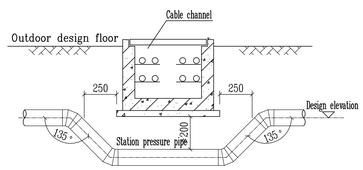
6）《电力建设施工及验收技术规范-管道篇》 DL5190.5-2012



Typical drawing of pipeline crossing(downwards)



Typical drawing of pipeline crossing(upwards)



Typical drawing of cable channel crossing(downwards)

Notes：

1、Pressure pipes above are make-up water pipe，industrial water pipe and potable water pipe.

2、Pipe crossing between water supply pipe & other pipes/trenches:

1）Avoid cable channel. 2）Avoid gravity drainage pipe.

3）Pipe with small diameter avoid pipe with large diameter.

3、Within the scope of roads,water supply pipe shall go under the roads.Within the scope of site, if water supply pipe shall go over the pipe,the space from top of pipe to floor shall be more than 500mm.

1、The station coordinate(AB) is adopted in this volume,station coordinate and UTM-43N system conversion formula refers to Civil Engineering volume.

2、EGM2008 elevation is adopted in this volume,unit of elevation is m.Average elevation of outdoor ground is 193.40m,and that of road and converter transformer square is 193.60m.

3. Unit of coordinate is m,that of dimensions is mm,and that of elevation and pipe elevation is m,elevation that is marked on pressure pipe represents elevation of pipe center.

4. Make-up water,industrial water & potable water source from underground water.Pipes are connected from oulet pipe of water treatment equipment area.Two industrial ponds and a comprehensive pump house are set in station area,with industrial pumps,frequency conversion water supply equipments and water tanks installed in comprehensive pump house..

Construction descriptions of pressure pipes in station area

5. Pipeline in this volume adopts directly buried installation,all piping should be layed on the surface of 100 mm thick coarse sand cushion, hydrostatic test shall be completed before installation,soil backfill shall be done after water pressure test and anticorrosive coating are qualified.Work pressure of industrial water pipe is 0.55 MPa,and that of test pressure is 1.05 MPa, the specific test requirements refer to《Code for construction and acceptance of water and sewerage pipeline works》(GB50268-2008).

6. Pipe backfill shall use original soil in principle,stones and bricks are not allowed,and compaction shall be done in layers at both sides simultaneously,dry density of backfilling sand shall reach 1.60t/m^3, that of backfilling soil shall be 1.45t/m^3, and dense degree shall be 0.93.

7. Load of road ground shall be designed according to the car-15 (aggravating double car tied).

8. Water in trench is not allowed for a long time during the whole installation process,measures shall be take to prevent floating pipe.

9. Industrial water pipe,potable waterpipe,make-up water pipe adopts galvanized steel pipe of lining plastic, with threaded connections.Piping and valves use flange connection.

10、Steel pipe anticorrosion and other basic requirements are as follows:

a. Surface pretreatment：surface of steel pipe shall be no oil contamination,soil,salinity,humidity,etc. Scale cinder and rust shall be removed thoroughly with sand blastor manual work，metallic lustre shall be apparent obviously,reaching Class Sa2.5 or St3 of ISO8501-1 standard,then anti-corrosion paint shall be coated immediately.In addition,surface of steel pipe should have a certain surface roughness,roughness shall be between the value of 32.5~75μm.

b. Painting environment: the surface of previous coating must be clear and dry before coating work, and besides,substrate temperature shall be 3℃ higher than dew point temperature. Relative humidity shall be less than 85%，in order to avoid affecting paint adhesion by condensation; Continuous environmental temperature shall be above 5℃,to guarantee curing reaction.

Position	Kinds	Name of paints	Layers	Thickness of each paint filmμm
Buried steel pipe (galvanized steel pipe) Ektexine	Confining bed	H53-42 closed epoxy coating	1	30
	Intermediate paint layer	842 epoxy micaceous iron oxide antirust paint	1	60
	Finishing coat	546 super high build epoxy tar antirust paint	2	125

Coating thickness and coating methods see explanatory demands of paint manufacturer. coating work shall be uniform coating,and not to leak,150 mm bare pipe shall be reserved at each end of pipe, in order to welding.

11. Unmarked way of pipeline crossing in this volume refer to this drawing, The crossing arrangement can be adjusted according to the need on site,corner shall adopt 45°elbows as far as possible,fitting materials are included in the length.

12. Pressure pipe shall comply with following standards and codes:

1）《Steel pipes of lining plastic for water supply》 CJ/T 136-2007

2）《Malleable iron threaded fitting of lining plastic for water supply》 CJ/T 137-2008

3）《Code for acceptance of construction quality of steel strucrures》 GB50205-2001

4）《The code of welding for power plant》 DL869-2012

5）《Code for construction and acceptance of water and sewerage pipeline works》 GB50268-2008

6）《Technical specification for thermal power erection and construction part 5:piping and system》 DL5190.5-2012

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - S G - 0 4 0 3 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed		
校核 Checked		
审核 Reviewed		
批准 Approved		
比例 Scale		图号 DWG NO.

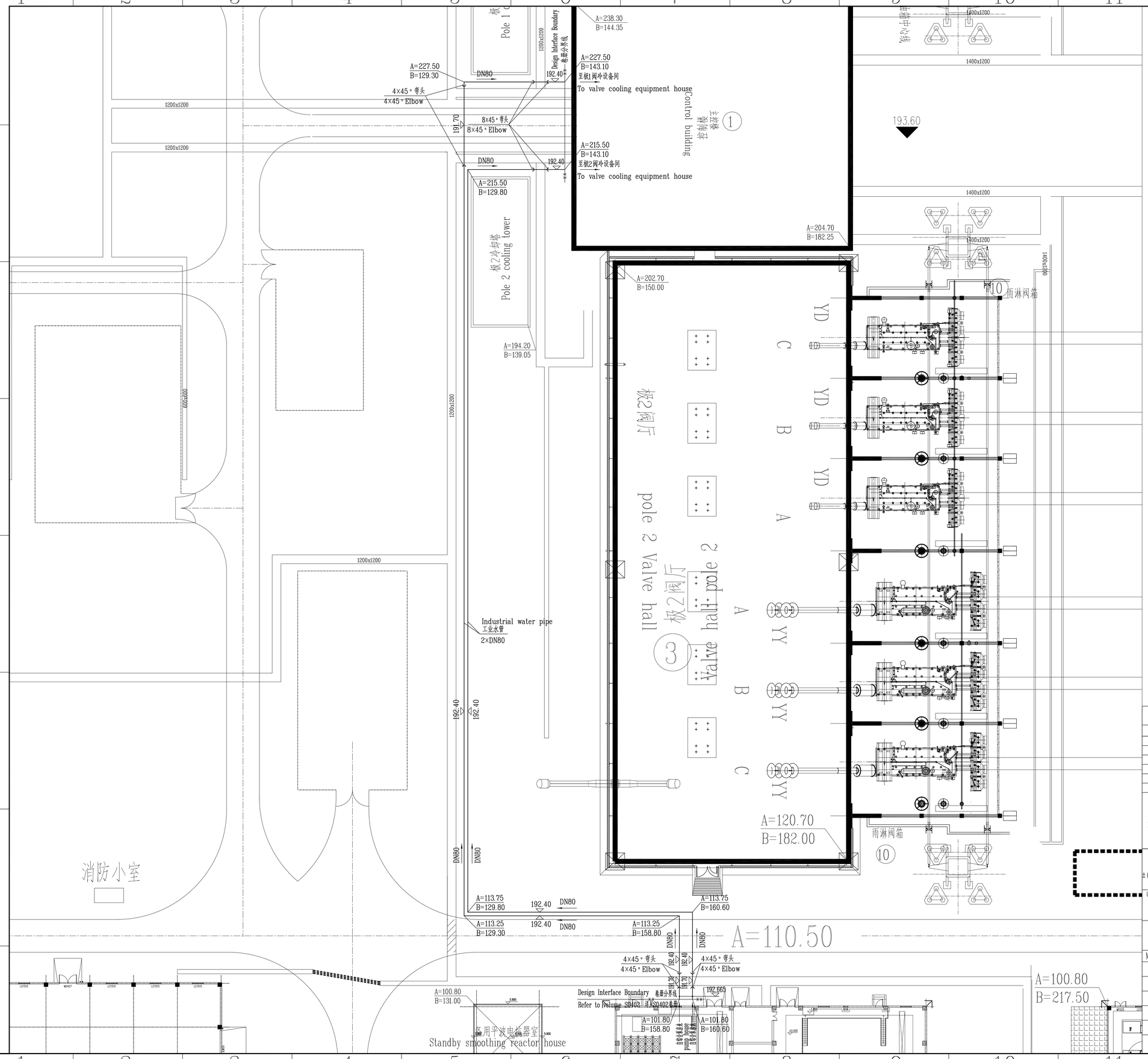
站区压力直埋管道施工说明

Description Of Construction For Buried Pressure Piping

30-B003701S-S0403-001

版本 REV. A

594X841



说明：

1. 本图卷册标高采用EGM2008高程系统，单位以m计。
2. 本卷册座标系统采用站区座标系统，单位以m计；站区座标系统与巴基斯坦UTM-43N平面坐标系的换算公式详见土建相关卷册。
3. 站内道路为郊区型道路。本图内道路、建筑物、构筑物相对位置仅为示意，以土建相关卷册为准。
4. 工业水管道防腐等施工说明、管沟交叉典型图详见本工程B003701S-S0403-01图《站区压力直埋管道施工说明》。
5. 本册图中管道均采用直埋敷设，所有管道应敷设在150mm厚中粗砂垫层上，安装完毕回填前进行严密性水压试验，水压试验合格后，再覆土回填。
6. 站区工业水管采用衬塑镀锌钢管，丝口连接，工作压力为0.55MPa，试验压力为1.05MPa，具体试验要求详见《给水排水管道工程施工及验收规范》(GB50268-2008)。
7. 管道施工、安装、回填、试压等按《给水排水管道工程施工验收规范》GB50268-2008规定执行。
8. 工业水水源从站区给水处理装置区域出水管接出。本工程设工业水池两座，容积均为1500m³。综合泵房一座，内设变频工业水泵4台(2用2备)，参数为Q=20m³/h，H=55m，N=5.5kW。

Notes:

1. EGM2008 elevation is adopted in this volume,unit of elevation is m.
2. The station coordinate(AB) is adopted in this volume,unit of coordinate is m; station coordinate(AB) and UTM-43N system conversion formula refers to Civil Engineering volume.
3. Station road adopts suburban road type.Relative location of roads, buildings and structures in the drawing is only for schematic drawing,details refer to Civil Engineering volume.
4. Construction description for pipe corrosion protection and pipeline crossing typical drawing refer to drawing B003701S-S0403-01《Construction description for directly buried pressure pipeline》.
5. Pipeline in this volume adopts directly buried installation,all piping should be layed on the surface of 150 mm thick coarse sand cushion, hydrostatic test shall be completed before installation,soil backfill shall be done afer water pressure is qualified.
6. Potable water pipeline uses galvanized steel pipe,with threaded connections.pressure is 0.55 MPa, test pressure is 1.05 MPa, the specific test requirements refer to《Code for construction and acceptance of water and sewerage pipeline works》(GB50268-2008).
7. Pipeline construction, installation, backfill and pressure test etc shall comply with《Code for construction and acceptance of water and sewerage pipeline works》(GB50268-2008).
8. Industrial water sources from outlet pipe of water treatment plant.Two industrial water ponds are set in station area,with a capacity of 2×1500m³.A comprehensive pump house is also located in station area, with 4 variable frequency pumps(2 in use,2 standby),parameter:Q=20m³/h, H=55m,N=5.5kW.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

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±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - S G - 0 4 0 3 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed		
校核 Checked		
审核 Reviewed		
批准 Approved		

站区工业水管道平面布置图

Industrial Water Piping Layout Drawing

比例 Scale	1:250	图号 DWG NO.	30-B003701S-S0403-002	版本 REV.	A

594X841

编号 No.	名 称 Name	规 格 Parameters	单位 Unit	数量	材料 Material	单重 Single (重量Weight)	总重 Total (重量Weight)	参考图号及备注 Remark
9								
8								
7								
6								
5								
4								
3	45° Elbow 45°弯头	DN80 PN=1.6Mpa	Set 套	12	Q235a lining plastic Q235a衬塑	-	-	DL/T 935-2005 钢塑复合管和管件 Pipe and fittings of steel lined with plastics
2	90° Elbow 90°弯头	DN80 PN=1.6Mpa	Set 套	6	Q235a lining plastic Q235a衬塑	-	-	DL/T 935-2005 钢塑复合管和管件 Pipe and fittings of steel lined with plastics
1	Steel Pipe lined with plastics 衬塑镀锌钢管	D89x4(DN100),PN=1.6Mpa	m 米	450	Q235a lining plastic Q235a衬塑	-	-	DL/T 935-2005 钢塑复合管和管件 Pipe and fittings of steel lined with plastics

编号 No.	名 称 Name	规 格 Parameters	单位 Unit	数量	材料 Material	单重 Single (重量Weight)	总重 Total (重量Weight)	参考图号及备注 Remark

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

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±660kV Lahore Converter Station

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M L H V D C - L C S - E C I - D D - S G - 0 4 0 3 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	甲	
校核 Checked		
审核 Reviewed		
批准 Approved		
比例 Scale		

站区工业水管材料表

Bill Of Material For Industrial Water Piping

图号 DWG NO.: 30-B003701C-S0403-003

版本 REV.: A

297X420